

-0.20	...	0.60
-1.10	...	-0.30
-0.70	...	1.20
1.00	...	0.80
-6.40	...	4.20
4.20	...	2.10
8.10	...	-0.30
-7.22	...	2.40

W

8-bit Quant W

$$\mathbf{W} = \mathbf{Q}_W^{(0)} \cdot s^{(0)}$$

$$\begin{aligned} \textcircled{1} \quad s^{(0)} &= \frac{|w|_{\max}}{q_{\max}} = \frac{|8.10|}{119} = 0.07 \\ \textcircled{2} \quad q_{w_{0,0}}^{(0)} &= \lfloor \frac{w_{0,0}}{s^{(0)}} \rfloor = \lfloor \frac{-0.20}{0.07} \rfloor = -3 \end{aligned}$$

-3	...	17
-16	...	-9
-10	...	34
15	...	23
-94	...	119
62	...	60
119	...	-9
-106	...	68
0.07	$s^{(0)}$	0.04

4-bit Quant $\mathbf{Q}_W$
(grouped)

$$\mathbf{Q}_{W_0}^{(0)} = (\mathbf{Q}_{W_0} - z_0) \cdot s_0^{(1)}$$

$$\mathbf{Q}_{W_1}^{(0)} = (\mathbf{Q}_{W_1} - z_1) \cdot s_1^{(1)}$$

$$\begin{aligned} \textcircled{1} \quad s^{(1)} &= \lfloor \frac{q_{w_{\max}}^{(0)} - q_{w_{\min}}^{(0)}}{q_{\max} - q_{\min}} \rfloor = \lfloor \frac{15 - -16}{15 - 0} \rfloor = 2 \\ \textcircled{2} \quad z &= \lfloor -\frac{q_{w_{\min}}^{(0)}}{s^{(1)}} \rfloor = \lfloor -\frac{-16}{2} \rfloor = 8 \\ \textcircled{3} \quad q_{w_{0,0}} &= \lfloor \frac{q_{w_{0,0}}^{(0)}}{s^{(1)}} + z \rfloor = \lfloor \frac{-3}{2} + 8 \rfloor = 7 \end{aligned}$$

7	...	9
0	...	0
3	...	14
15	...	11
8	z_0	3
2	$s_0^{(1)}$	3
1	...	14
11	...	8
15	...	0
0	...	9
7	z_1	1
15	$s_1^{(1)}$	9
0.07	$s^{(0)}$	0.04

$$\hat{q}_{w_{0,0}}^{(0)} = (q_{w_{0,0}} - z) \cdot s^{(1)} = (7 - 8) \cdot 2 = -2$$

Dequant $\mathbf{Q}_W^{(0)}$
(grouped)

-2	...	18
-16	...	-9
-10	...	33
14	...	24
-90	...	117
60	...	63
120	...	-9
-105	...	72

Within
INT8 Range

INT8
Computation